Highlights

Dual-Pathway Deep Learning for Power Transformer Vibration Fault Diagnosis: Spectral Feature Encoding and Spatial Variational Autoencoder

- Validated on 19 real in-service transformers with strict cross-unit test split (no unit in more than one split).
- SpecRas spectral rasterisation with Grad-CAM yields Hz-resolution spectral attribution maps.
- HetSpatVAE detects anomalies label-free via spatial reconstruction residuals over the time-frequency plane.
- SWA adds +2.75 pp test accuracy with zero missed faults under cross-unit covariate shift.
- Peak-load false positives suggest load metadata as the primary missing covariate.

Dual-Pathway Deep Learning for Power Transformer Vibration Fault Diagnosis: Spectral Feature Encoding and Spatial Variational Autoencoder

Abstract

Vibration-based fault diagnosis for in-service power transformers faces three compounding challenges: non-stationary harmonic spectra under varying load, severe label scarcity (a handful of confirmed faults per unit per decade), and the need for frequency-level interpretability before engineers can justify maintenance decisions.

This paper presents a dual-pathway deep learning framework validated on 19 real in-service alternating current (AC) power transformers, partitioned by unit identity (15 training / 2 validation / 2 held-out test; 400 test segments: 246 normal, 154 fault). The supervised SpecRas (Spectral Rasterisation) pathway rasterises a 1,200-dimensional spectral feature vector—time-domain statistics, Welch power spectral density (PSD) at 1–2,000 Hz, and high-frequency energy ratios—into a 150×150 image and fine-tunes a ResNet18 classifier with stochastic weight averaging (SWA); Gradient-weighted Class Activation Mapping (Grad-CAM) converts predictions to per-Hz spectral attribution maps. The HetSpatVAE (Heterogeneous Spatial Variational Autoencoder) pathway encodes three-channel time-frequency images (Morlet continuous wavelet transform (CWT), short-time Fourier transform (STFT), waveform context) through a spatial ResNet variational autoencoder; reconstruction residuals localise anomalies without fault labels. The *AI contributions* are spectral feature rasterisation, a spatial VAE architecture, and SWA-based training; the *engineering application* is interpretable fault diagnosis on in-service power transformers under cross-unit covariate shift.

On 400 held-out samples from two previously unseen transformers (246 normal, 154 fault), SpecRas achieves **97.00% accuracy** (macro-F1 = 0.969, AUC = 0.989) with zero missed faults; HetSpatVAE reaches **98.25% accuracy** (macro-F1 = 0.982) with zero missed faults and just 7 false positives. All false positives cluster in the evening peak-load window with elevated 1,000–2,000 Hz STFT content, consistent with load as a missing covariate.

Keywords: condition monitoring, deep learning, industrial AI, transformer fault diagnosis, variational autoencoder, vibration signal analysis

1. Introduction

A failed AC power transformer can black out a region for hours; replacement takes months to years because each unit is largely custom-built (CIGRE Working Group A2.37, 2015). Knowing whether a unit is degrading—before it fails—is therefore worth considerable effort. Surface-mounted accelerometers have become the standard non-intrusive monitoring modality: they can be retrofitted without de-energising the unit and are sensitive to core looseness and winding deformation at a fraction of the cost of dissolved gas analysis (DGA) oil sampling (Bagheri et al., 2018; Tenbohlen et al., 2016).

Three practical constraints make vibration-based transformer diagnosis harder than the bearing-fault literature suggests.

Non-stationarity. Transformer vibration stems from two coupled mechanisms: magnetostriction-induced Lorentz forces at even harmonics of the supply frequency (100, 200 Hz for a 50 Hz system), and load-induced winding forces at the same harmonics plus a broadband floor (Zollanvari et al., 2021). Both shift continuously with operating conditions. A fault redistributes energy across frequency decades—but so does a routine load step, and the two look similar in a fixed-window Fourier representation (Hong et al., 2021).

Label scarcity. A large regional grid may see only a handful of confirmed fault cases per transformer per decade. Large, balanced, annotated training sets are simply not available, and accumulating matched fault recordings for every unit is impractical (Zhao et al., 2020).

Interpretability. Grid engineers are accountable for every maintenance decision. An anomaly score alone cannot justify taking a high-value asset offline; the model must indicate which frequency bands drove the conclusion so engineers can cross-check against known physical fault signatures (Rudin, 2019).

Limitations of existing approaches. Rule-based methods encode domain expertise as spectral thresholds and are fully transparent, but their accuracy is bounded by calibration conditions and they cannot adapt to unseen fault morphologies (IEEE Power and Energy Society, 2019). Empirical mode decomposition (EMD) and wavelet decomposition track non-stationarity better but require manual feature selection (Peng and Chu, 2004). Convolutional neural network (CNN) classifiers on spectrograms achieve high accuracy on rotating-machinery benchmarks but require balanced labelled data and produce no frequency-level explanation (Guo et al., 2019; Wen et al., 2018). Flat-vector variational autoencoder (VAE) anomaly detectors avoid label requirements but cannot localise which region of the time-frequency image is anomalous (Kingma and Welling, 2014; An and Cho, 2015).

Contributions. This paper makes four technical contributions (Fig. 1).

1. The SpecRas (Spectral Rasterisation) pathway rasterises a 1,200-dimensional spectral feature vector (time-domain statistics, Welch power spectral density (PSD) at 1–2 000 Hz, and high-frequency energy ratios) into a 150×150 image whose PSD-dominant strip preserves frequency ordering. Following the feature-to-image encoding of Zerone (Kim and Yoon, 2025) and substituting physics-motivated spectral descriptors for transformer vibration, Gradient-weighted Class Activation Mapping (Grad-CAM) activations in the PSD rows read directly as per-Hz spectral attribution maps.
2. The HetSpatVAE (Heterogeneous Spatial Variational Autoencoder) pathway encodes a three-channel heterogeneous time-frequency image into a 64×7×7 latent map via a Spatial ResNet VAE (SpatialResNetVAE); each spatial cell corresponds to a specific time-frequency region, so reconstruction residuals serve as interpretable anomaly heat maps.
3. Stochastic weight averaging (Izmailov et al., 2018) contributes +2.75 pp test accuracy over the best-validation checkpoint on the 3,844-sample training pool under cross-unit covariate shift, with zero missed faults.
4. A failure-mode analysis reveals that all false positives from both pathways cluster in the 16:00–20:30 peak-load window and carry elevated 1 000–2 000 Hz short-time Fourier transform (STFT) content, pointing to load metadata conditioning as a concrete next step.

Section 2 surveys related work. Section 3 describes the methodology. Section 4 reports experiments and discussion. Section 5 concludes.

2. Related Work

2.1. Vibration-Based Transformer Fault Diagnosis

Transformer condition monitoring has traditionally centred on dissolved gas analysis and power-frequency electrical tests (IEEE Power and Energy Society, 2019; CIGRE Working Group A2.37, 2015), both requiring oil sampling or controlled energisation. Vibration analysis emerged as a non-intrusive complement (Bagheri et al., 2018): core looseness, winding deformation, and cooling faults each perturb the 100–200 Hz harmonic components in recognisable ways. Early systems extracted these features by hand and applied fixed thresholds or classical classifiers (Tenbohlen et al., 2016; Zollanvari et al., 2021). More recent work uses deep recurrent networks on waveform windows and CNN classifiers on vibration spectrograms (Hong et al., 2021; Li et al., 2023), with reported accuracies above 90% on within-unit test sets. Cross-unit evaluation—where test transformers are entirely absent from training—is less commonly reported and consistently harder.

2.2. Deep Learning for Industrial Condition Monitoring

CNN classifiers on spectrograms are now standard for rotating-machinery fault diagnosis, reaching near-perfect accuracy on benchmarks such as Case Western Reserve University (CWRU) and Jiangnan University (JNU) benchmarks (Guo et al., 2019; Wen et al., 2018). Attention mechanisms and recurrent architectures extend the temporal modelling horizon (Zhao et al., 2019, 2017). Transfer learning from ImageNet cuts the labelled-data requirement substantially (Shao et al., 2019); deep residual shrinkage networks add robustness to noisy labels (Zhao et al., 2020). Power transformer vibration differs from bearing diagnostics in two ways. The training set is far smaller—hundreds of samples per class, not thousands. And the signal concentrates in narrow harmonic peaks at multiples of the supply frequency, so raw-signal CNNs without explicit frequency-domain feature engineering tend to miss the discriminative structure.

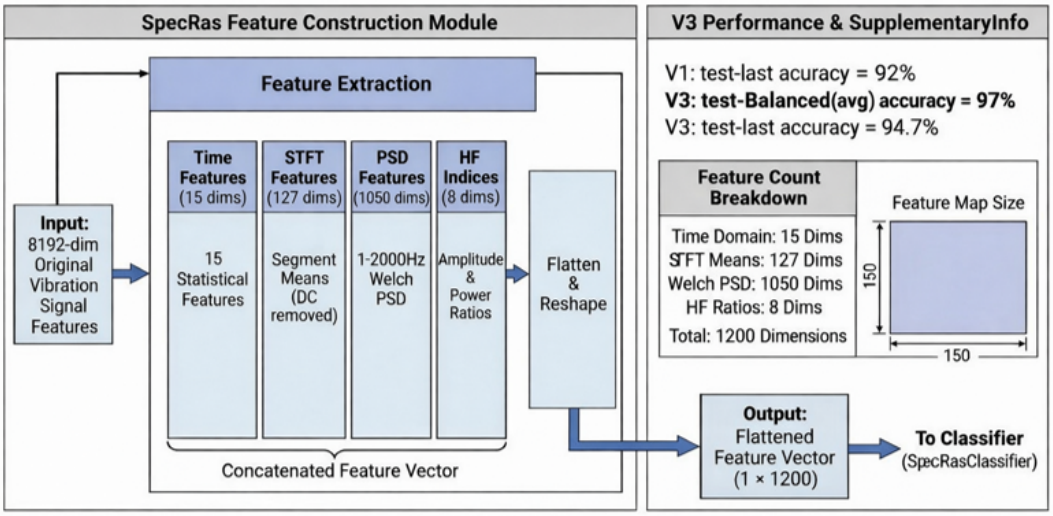
2.3. Signal Representation and Feature Learning

How a signal is represented before it reaches a model has a larger effect on diagnostic accuracy than the model architecture itself, particularly when training data are limited. Continuous wavelet transform (CWT) resolves transient events at multiple time-frequency scales and has been standard for transformer fault detection (Peng and Chu, 2004). STFT is preferred for quasi-stationary harmonic signals because of its computational efficiency and fixed spectral resolution (Cohen, 1995). Combining CWT and STFT captures complementary fault signatures (Lei et al., 2020). Signal-to-image encoding for CNN processing ranges from raw waveform reshaping (Wen et al., 2018) to stacked short-time spectra as image channels (Zhao et al., 2019). End-to-end 1-D CNNs learn task-relevant frequency filters without manual selection (Guo et al., 2019). Our SpecRas encoding adapts the feature-to-image rasterisation principle of Zerone (Kim and Yoon, 2025) to physics-motivated spectral features specific to power transformer vibration. Rather than mapping binary or generic indicators, SpecRas encodes a 1,200-dimensional spectral feature vector (Welch PSD at 1–2 000 Hz, STFT frame means, and high-frequency energy ratios) into a physically ordered raster image. The PSD panel occupies ≈77% of image height with frequency ordering preserved row-by-row, so Grad-CAM activations read directly as Hz-resolution spectral attribution maps.

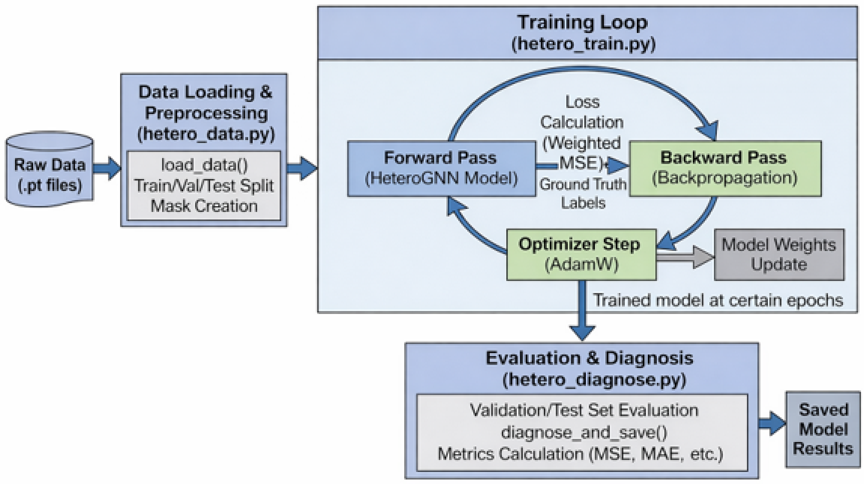
2.4. Generative Models for Anomaly Detection

VAEs (Kingma and Welling, 2014) are well suited to one-class anomaly detection: a model trained exclusively on normal samples produces high reconstruction error and out-of-distribution latent codes for anomalous inputs (An and Cho, 2015). The β-VAE (Higgins et al., 2017) promotes disentangled representations at the cost of reconstruction fidelity. Deep Autoencoding Gaussian Mixture Models (Zong et al., 2018) combine reconstruction error with latent-space density estimation; PatchCore (Roth et al., 2022) builds a memory bank of normal feature patches for industrial image anomaly detection. Both use flat vector latent codes and therefore cannot say where in the time-frequency image the anomaly occurred. HetSpatVAE addresses this by retaining the $7{\times}7$ spatial structure of the ResNet encoder, with posterior collapse managed via β-KL warm-up annealing (Lucas et al., 2019).

3. Methodology



(a) SpecRas supervised pathway: 1 200-dim spectral feature vector → 150 × 150 raster image → ResNet18 + SWA. Best-val checkpoint (V1) reaches 94.25%; SWA checkpoint (V3) reaches 97.00%.



(b) HetSpatVAE one-class pathway: data preprocessing → SpatialResNet-VAE training → composite anomaly scoring. Trained on normal samples only; achieves 98.25% accuracy.

Figure 1: Dual-pathway vibration fault diagnosis framework. **(a)** SpecRas supervised pathway: spectral rasterisation → ResNet18 + SWA → per-Hz Grad-CAM attribution. **(b)** HetSpatVAE one-class pathway: 3-channel time-frequency image → SpatialResNetVAE → reconstruction-residual heat map. Both branches share energy-weighted multi-channel fusion (Section 3.2).

3.1. Problem Formulation

Let $\mathcal{D} = \{(\mathcal{X}_k, y_k)\}_{k=1}^{N}$ be a labelled dataset of vibration samples, where

$$\mathcal{X}_k = \left\{ x_k^{(u)} \in \mathbb{R}^T \;\middle|\; u = 1, \ldots, U_k \right\} \tag{1}$$

is the set of multi-channel signals acquired at timestamp t_k, $T = 8{,}192$ is the fixed signal length (sampling rate $f_s = 8{,}192\,\mathrm{Hz}$, 1-second window), $U_k \geq 1$ is the number of active sensor channels, and $y_k \in \{0, 1\}$ is the binary label ($0 =$ normal, $1 =$ fault). The diagnostic task has two objectives: (i) supervised binary classification given $\mathcal{X}_k$ and a labeled training set; and (ii) one-class anomaly detection given only normal-class training data via a composite scoring function. Both require interpretable evidence linking the decision to specific frequency bands.

3.2. Signal Preprocessing

Energy-weighted channel fusion. To collapse $\mathcal{X}_k$ into a single representative signal, the channel weights proportional to the signal energy are calculated:

$$w_u = \frac{\mathbb{E}\Big[(x_k^{(u)})^2\Big]}{\sum_{j=1}^{U_k} \mathbb{E}\Big[(x_k^{(j)})^2\Big] + \varepsilon}, \tag{2}$$

$$\tilde{x}_k = \sum_{u=1}^{U_k} w_u\, x_k^{(u)}.$$

with $\varepsilon = 10^{-12}$. This fusion is invariant to channel ordering and count, so all downstream encoders remain agnostic to the sensor configuration of each individual transformer. The acquired JSON records contain only a `sensor_id` string and signal samples (no position, mounting angle, or sensor type is recorded), so energy-weighted aggregation is the natural choice for producing a fixed-size representation regardless of how many channels happen to be active.

DC removal and label assignment. Both pathways operate on the mean-subtracted signal $\bar{x}_k = \tilde{x}_k - \mu_{\tilde{x}_k}$. Class labels are inferred automatically from directory names: subdirectories containing normal receive label 0; those containing fault receive label 1.

3.3. Supervised SpecRas–ResNet18 Pathway

3.3.1. Spectral Feature Vector

For each DC-removed channel signal $\bar{x}^{(u)} \in \mathbb{R}^T$, a $d = 1{,}200$-dimensional characteristic vector is extracted following the schema in Table 1.

Table 1: SpecRas feature schema ($d = 1{,}200$ total dimensions). Multi-channel results are aggregated by energy-weighted mean (Eq. 2).

Segment	Method	Dims	Description
Time	Statistics	15	Mean, RMS, variance, standard deviation, maximum, minimum, peak-to-peak, kurtosis, skewness, zero-crossing rate, mean absolute value, crest factor, impulse factor, margin factor, and waveform factor
STFT	Frame means	127	Mean magnitude per frame (n_{seg}=128, hop=64, DC bin removed)
PSD	Welch	1 050	1–1 000 Hz at 1 Hz resolution (1 000 bins) + 1 001–2 000 Hz in 20 Hz bands (50 bins)
HF	Ratios	8	Amplitude ratio and power ratio above each of $f_{\text{thr}} \in \{1000, 2000, 3000, 4000\}$ Hz
Total		**1 200**	

Time-domain features. For signal $\bar{x} \in \mathbb{R}^T$, the 15 time-domain statistics are

$$\begin{aligned}\mathbf{z}_{\text{time}} = \big[&\mu,\ x_{\text{RMS}},\ \sigma^2,\ \sigma,\ x_{\max},\\ &x_{\min},\ x_{\text{p2p}},\ \kappa,\ \gamma,\ r_{\text{zc}},\\ &\mu_{|\cdot|},\ x_{\text{crest}},\ x_{\text{imp}},\\ &x_{\text{margin}},\ x_{\text{wave}}\big] \in \mathbb{R}^{15}.\end{aligned} \tag{3}$$

Let

$$\begin{aligned}\mu &= T^{-1}\sum_{t=1}^{T}\bar{x}_t,\\ \sigma^2 &= T^{-1}\sum_{t=1}^{T}(\bar{x}_t-\mu)^2,\\ \sigma &= \sqrt{\sigma^2}.\end{aligned} \tag{4}$$

Then the main quantities are defined as

$$
\begin{aligned}
x_{\mathrm{RMS}} &= \sqrt{T^{-1}\sum_{t=1}^{T}\bar{x}_t^2}, \\
x_{\max} &= \max_t \bar{x}_t, \qquad x_{\min} = \min_t \bar{x}_t, \\
x_{\mathrm{p2p}} &= x_{\max} - x_{\min}, \\
\kappa &= \frac{m_4}{m_2^2}, \\
\gamma &= \frac{m_3}{\sigma^3}, \\
r_{\mathrm{zc}} &= \frac{1}{T-1}\sum_{t=1}^{T-1}\mathbf{1}[\bar{x}_t\,\bar{x}_{t+1} < 0], \\
\mu_{|\cdot|} &= T^{-1}\sum_{t=1}^{T}|\bar{x}_t|, \\
x_{\mathrm{crest}} &= \frac{x_{\max}}{x_{\mathrm{RMS}}}, \\
x_{\mathrm{imp}} &= \frac{x_{\max}}{\mu_{|\cdot|}}, \\
x_{\mathrm{wave}} &= \frac{x_{\mathrm{RMS}}}{\mu_{|\cdot|}}, \\
x_{\mathrm{margin}} &= \frac{x_{\max}}{\left(T^{-1}\sum_{t=1}^{T}|\bar{x}_t|^4\right)^{1/4}}.
\end{aligned}
\tag{5}
$$

where $m_p = T^{-1}\sum_{t=1}^{T}(\bar{x}_t - \mu)^p$ denotes the p-th central moment.

STFT features. After DC removal, a short-time Fourier transform is applied with Hann window, segment length $n_{\mathrm{seg}} = 128$, and overlap $n_{\mathrm{overlap}} = 64$. Let $S(f, \tau)$ denote the complex STFT of $\bar{x}$. After discarding the DC bin, we compute the mean magnitude of each time frame as

$$
s_\tau = \frac{1}{|\mathcal{F}'|}\sum_{f\in\mathcal{F}'}|S(f,\tau)|, \tag{6}
$$

where $\mathcal{F}'$ is the non-DC frequency index set. The resulting sequence is zero-padded or truncated to 127 dimensions, yielding a fixed-length STFT descriptor

$$
\mathbf{s}_{\mathrm{STFT}} \in \mathbb{R}^{127}. \tag{7}
$$

PSD features. Welch PSD is estimated at 1 Hz resolution from 1 to 2000 Hz. The low-frequency region 1–1000 Hz is retained at full resolution, while the high-frequency region 1001–2000 Hz is compressed into 20 Hz bands by local averaging. This gives

$$
\begin{aligned}
\mathbf{p}_{\mathrm{PSD}} = \big[&P(1),\, P(2),\, \ldots,\, P(1000), \\
&\bar{P}_{1001:1020},\, \ldots,\, \bar{P}_{1981:2000}\big] \in \mathbb{R}^{1050}.
\end{aligned}
\tag{8}
$$

High-frequency ratio features. To explicitly characterise the proportion of elevated-frequency content, we define four threshold frequencies

$$
f_{\mathrm{thr}} \in \{1000, 2000, 3000, 4000\}\ \mathrm{Hz}.
$$

For each threshold, two scalar ratios are computed: the amplitude ratio

$$r^{(A)}(f_{\mathrm{thr}}) = \frac{\sum_{f \geq f_{\mathrm{thr}}} A(f)}{\sum_{f \geq 0} A(f)}, \tag{9}$$

and the power ratio

$$r^{(P)}(f_{\mathrm{thr}}) = \frac{\sum_{f \geq f_{\mathrm{thr}}} P(f)}{\sum_{f \geq 0} P(f)}, \tag{10}$$

where $A(f)$ and $P(f)$ denote the spectrum magnitude and PSD, respectively. Concatenating the four threshold pairs yields

$$\begin{aligned} \mathbf{r}_{\mathrm{HF}} = \big[&r^{(A)}(1000),\, r^{(P)}(1000),\\ &r^{(A)}(2000),\, r^{(P)}(2000),\\ &r^{(A)}(3000),\, r^{(P)}(3000),\\ &r^{(A)}(4000),\, r^{(P)}(4000)\big] \in \mathbb{R}^8. \end{aligned} \tag{11}$$

Multi-channel aggregation. For a sample with U available sensor channels, each channel yields a feature vector $\mathbf{z}_u \in \mathbb{R}^{1200}$. To preserve the relative importance of stronger vibration responses, the sample-level feature is computed by an energy-weighted mean:

$$\mathbf{z} = \frac{\sum_{u=1}^{U} e_u \mathbf{z}_u}{\sum_{u=1}^{U} e_u + \varepsilon}, \tag{12}$$

where $e_u = T^{-1} \sum_{t=1}^{T} x_{u,t}^2$ is the channel energy and ε is a small constant for numerical stability.

3.3.2. Raster Stripe Image Encoding

The aggregate feature vector $\mathbf{S}_k$ is mapped to a 150×150 RGB image using a non-interpolating raster layout (Fig. 2). Features are tiled into a 2D grid with unit cell size $W_{\mathrm{unit}} \times H_{\mathrm{unit}} = 3 \times 2$ pixels, and the four panels are stacked vertically in order (time / STFT / PSD / HF), with each panel wrapping its features row-by-row at a fixed tile width.

$$n_{\mathrm{rows}}^{(g)} = \lceil d_g / w_g \rceil, \tag{13}$$

where $g \in \{\mathrm{time}, \mathrm{stft}, \mathrm{psd}, \mathrm{hf}\}$, d_g is the segment dimension, and w_g is the tile width (16 / 16 / 32 / 8 for the four panels). The PSD panel spans 33 rows $\times$ 2 px = 66 px, against 2 px for time and HF and 16 px for STFT, so it occupies approximately 77% of the image content area.

Pixel values are linearly normalised using training-set statistics:

$$I_{ij} = \mathrm{clip}\left(\frac{S_d - \hat{m}_d}{\hat{M}_d - \hat{m}_d + \varepsilon},\ 0, 1\right), \tag{14}$$

where $\hat{m}_d$ and $\hat{M}_d$ are training-set minimum and maximum for dimension d. The image is padded to 150×150 and represented as a three-channel input tensor for ResNet18-based classification.

Two properties of this layout matter for interpretability and learning. Each PSD row holds 32 tiles spanning 32 Hz: row r covers $1 + 32r$ to $32(r+1)$ Hz, so Grad-CAM activations in those rows read directly as frequency bands. The 77% height share also concentrates most CNN receptive fields on spectral content; the time and HF panels occupy narrow border strips.

3.3.3. ResNet18 Training and SWA

A ResNet18 (He et al., 2016) classifier is trained from scratch on SpecRas images. The final fully-connected layer is replaced with a two-output linear head ($\mathbb{R}^{512} \to \mathbb{R}^2$). Table 2 lists the complete hyperparameters.

Table 2: Training configuration for SpecRas ResNet18. All values were fixed before any test-set evaluation.

Hyperparameter	Value
Image size	150×150
Batch size	16
Max epochs	30
Base learning rate	8×10^{-6}
LR schedule	ReduceLROnPlateau (val macro-F1)
Weight decay	10^{-4}
Label smoothing	$\varepsilon = 0.05$
Class weights	Inverse class frequency
Gradient clipping	$\|\mathbf{g}\|_2 \leq 1.0$
EMA decay	0.999
SWA start ratio	0.50 of max epochs
SWA min-LR	5×10^{-5}
SWA max-loss bias	0.05 above best val loss
Random seed	42

Loss function. Cross-entropy with label smoothing and class weights:

$$\mathcal{L}_{\mathrm{CE}} = -\sum_{c=0}^{1} \alpha_c \left[(1-\varepsilon)\, y_c + \tfrac{\varepsilon}{2}\right] \log \hat{p}_c, \tag{15}$$

where $\alpha_c \propto n_c^{-1}$ are inverse-frequency class weights.

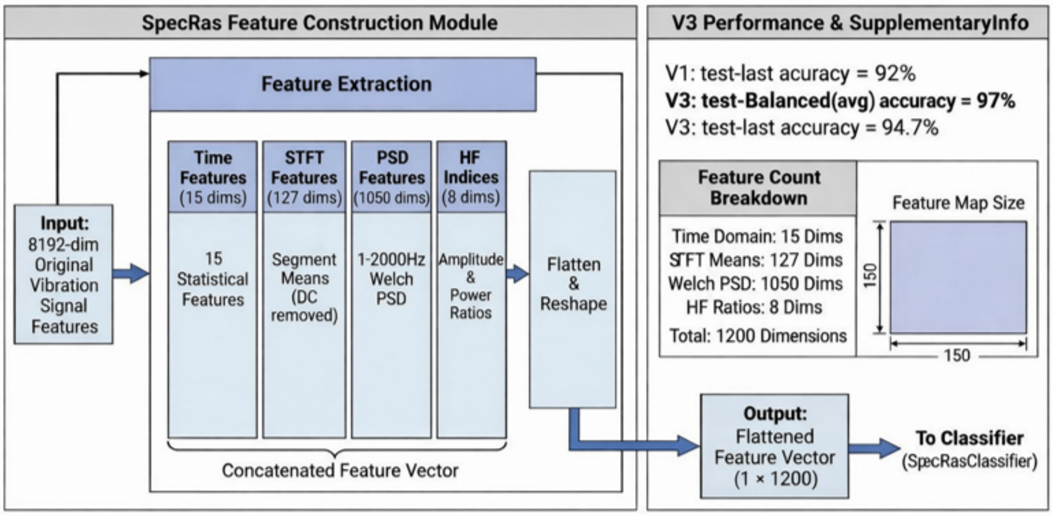
Model selection. The best checkpoint is selected by the four-key tie-break

$$\big(F1_{\mathrm{val}}\!\uparrow,\ L_{\mathrm{val}}\!\downarrow,\ \mathrm{lr}\!\downarrow,\ \mathrm{epoch}\!\uparrow\big).$$

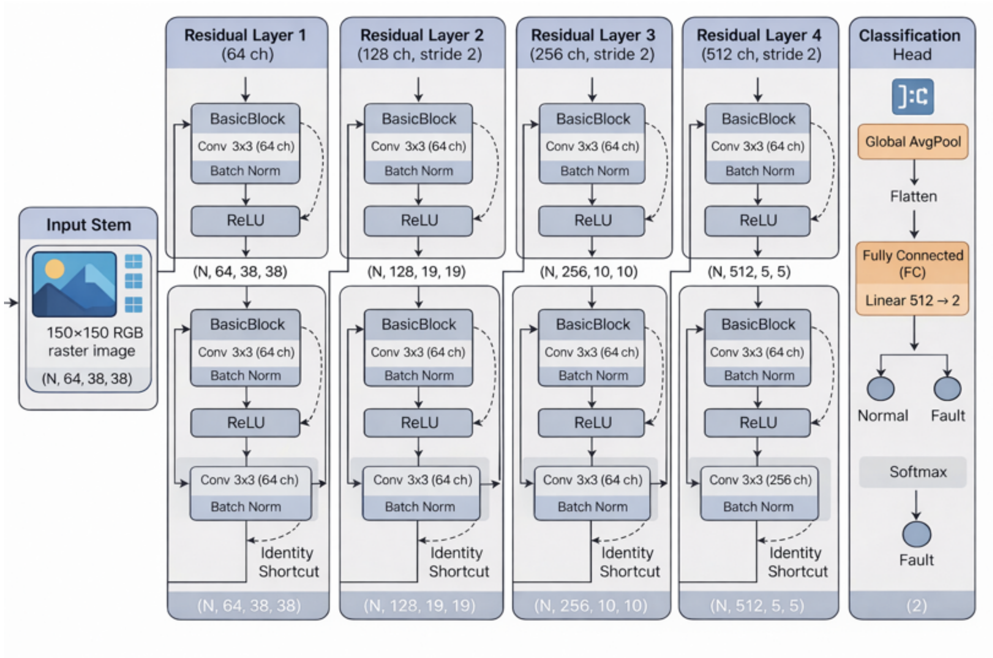
An EMA copy (decay 0.999) of the parameters runs in parallel.

Stochastic weight averaging. In the second half of training, whenever the current validation loss falls within 5% of the best observed and the learning rate drops below a threshold, the weights are accumulated into an SWA buffer (Izmailov et al., 2018). After training, batch-normalisation running statistics are re-computed on 128 forward passes.

Grad-CAM attribution. Grad-CAM (Selvaraju et al., 2017) is applied to the final residual block (`layer4`). The resulting 5×5 activation map is upsampled to 150×150 and overlaid on the raster image. Hot regions within the PSD panel (rows 1–33 of that panel) correspond to specific frequency bands: a pixel at in-panel row r (0-based), column c maps to $f = 1 + 32r + c$ Hz, giving per-Hz spectral attribution without additional decoding.



(a) Feature construction: 4 descriptor groups (time 15 + STFT 127 + PSD 1 050 + HF 8 = 1 200 dims) → 150 × 150 raster image. PSD occupies ≈77% of the image; each PSD row spans 32 Hz so Grad-CAM activations read directly as Hz bands.



(b) ResNet18 backbone: four residual layers (64 → 128 → 256 → 512 channels), global average pooling, two-class linear head ($\mathbb{R}^{512} \to \mathbb{R}^2$).

Figure 2: SpecRas pipeline. **(a)** Feature extraction and raster encoding: PSD dominates 77% of the 150 × 150 image; Grad-CAM hot rows map directly to 1–1 000 Hz spectral bands. **(b)** ResNet18 classifier backbone.

3.4. HetSpatVAE Anomaly Detection Pathway

3.4.1. Heterogeneous Three-Channel Encoding

The energy-weighted z-score normalised signal $\check{x} = (\tilde{x} - \mu)/\sigma$ is converted to a three-channel 224×224 image $\mathbf{I} \in [0, 1]^{3\times224\times224}$. Let

$$\mathbf{M}_0 = \left|\mathrm{CWT}(\check{x},\ \Psi_{\mathrm{morl}},\ a_{1:128})\right|, \tag{16}$$
$$\mathbf{M}_1 = \left|\mathrm{STFT}(\check{x};\ n{=}256,\ h{=}128)\right|, \tag{17}$$
$$\mathbf{M}_2 = \mathrm{fold}(\check{x},\ 32\times256), \tag{18}$$

where scales $a_{1:128}$ are logarithmically spaced and fold(·) reshapes the 1-D signal into a 32×256 matrix. Each channel is then:

$$\mathbf{I}_{[n]} = \mathrm{norm}\big(\mathrm{resize}(\mathbf{M}_n,\ 224\times224)\big), \quad n = 0, 1, 2. \tag{19}$$

The three channels provide complementary physical views: CWT resolves transient signatures at multiple time-frequency scales; STFT reveals stationary harmonic structure; context encodes global waveform morphology.

3.4.2. SpatialResNetVAE Architecture

Encoder. A ResNet18 backbone processes $\mathbf{I}$ through its stem and four residual blocks, producing a feature tensor $\mathbf{h} \in \mathbb{R}^{512\times7\times7}$. Two parallel 1×1 convolutional heads project $\mathbf{h}$ to spatial mean and log-variance, both in $\mathbb{R}^{64\times7\times7}$:

$$\begin{aligned} \boldsymbol{\mu}_z &= \mathbf{W}_\mu * \mathbf{h}, \\ \log \boldsymbol{\sigma}_z^2 &= \mathbf{W}_\sigma * \mathbf{h}. \end{aligned} \tag{20}$$

Retaining the 7×7 spatial dimensions (rather than global average pooling) is the single most important architectural decision: each spatial cell (j, k) has a 32×32-pixel receptive field in $\mathbf{I}$, so reconstruction residuals at cell (j, k) correspond to a specific time-frequency region of the input CWT or STFT.

Reparameterisation. $\mathbf{z} = \boldsymbol{\mu}_z + \boldsymbol{\sigma}_z \odot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Decoder. A 1×1 convolution maps $\mathbf{z} \in \mathbb{R}^{64\times7\times7}$ to 512 channels; five transposed-convolution blocks (ConvTranspose2d, k=4, s=2, p=1, BatchNorm, ReLU) progressively upsample to 3×224×224. The output activation is Sigmoid.

Training objective. The β-VAE evidence lower bound (ELBO) (Kingma and Welling, 2014; Higgins et al., 2017) with L1 reconstruction:

$$\begin{aligned} \mathcal{L} = &\frac{1}{B}\,\|\mathbf{I} - \hat{\mathbf{I}}\|_1 \\ &+\ \beta(e) \cdot \frac{-1}{2B} \sum_{i,j,k} \big(1 + \log \sigma_{z,ijk}^2 - \mu_{z,ijk}^2 - \sigma_{z,ijk}^2\big), \end{aligned} \tag{21}$$

where $\beta(e) = \min(\beta_{\max}, \beta_{\max} \cdot e/e_{\mathrm{warm}})$ anneals from 0 to $\beta_{\max} = 0.05$ over $e_{\mathrm{warm}} = 20$ epochs. L1 reconstruction preserves edge sharpness in the CWT scalogram channel; the Kullback–Leibler (KL) warm-up prevents posterior collapse (Lucas et al., 2019). Training uses only normal-labelled samples.

3.4.3. Composite Anomaly Score

Channel-weighted reconstruction error.

$$
\begin{aligned}
e_{\text{rec}}(\mathbf{I}) = 0.4\,\|\mathbf{I}_{[0]} - \hat{\mathbf{I}}_{[0]}\|_1 \\
+ 0.5\,\|\mathbf{I}_{[1]} - \hat{\mathbf{I}}_{[1]}\|_1 \\
+ 0.1\,\|\mathbf{I}_{[2]} - \hat{\mathbf{I}}_{[2]}\|_1.
\end{aligned} \tag{22}
$$

The STFT channel receives the highest weight (0.5) because structured harmonic patterns are most sensitive to fault-induced changes.

Mahalanobis distance. The spatial mean of the encoder output $\bar{\boldsymbol{\mu}}_z = \frac{1}{49}\sum_{j,k}\boldsymbol{\mu}_{z,\cdot jk} \in \mathbb{R}^{64}$ provides a fixed-length embedding. A Gaussian density is fitted to all training-set embeddings:

$$\boldsymbol{\mu}_{\text{tr}} = N^{-1}\textstyle\sum_n \bar{\boldsymbol{\mu}}_{z,n}, \tag{23}$$

$$\boldsymbol{\Sigma}_{\text{tr}} = N^{-1}\textstyle\sum_n (\bar{\boldsymbol{\mu}}_{z,n} - \boldsymbol{\mu}_{\text{tr}})(\bar{\boldsymbol{\mu}}_{z,n} - \boldsymbol{\mu}_{\text{tr}})^\top + \varepsilon\mathbf{I}, \tag{24}$$

$$d_{\text{MD}}(\mathbf{I}) = \sqrt{(\bar{\boldsymbol{\mu}}_z - \boldsymbol{\mu}_{\text{tr}})^\top \boldsymbol{\Sigma}_{\text{tr}}^{-1}(\bar{\boldsymbol{\mu}}_z - \boldsymbol{\mu}_{\text{tr}})}. \tag{25}$$

Composite score. Both scores are z-normalised using training statistics, then combined:

$$
\begin{aligned}
a(\mathbf{I}) = 0.6 \cdot \frac{e_{\text{rec}} - \mu_{\text{rec}}}{\sigma_{\text{rec}}} \\
+ 0.4 \cdot \frac{d_{\text{MD}} - \mu_{\text{MD}}}{\sigma_{\text{MD}}}.
\end{aligned} \tag{26}
$$

A sample is flagged when $a(\mathbf{I}) \geq \tau$, with τ at the 97.5th percentile of training normal scores. Table 3 lists all HetSpatVAE hyperparameters.

Table 3: HetSpatVAE training hyperparameters.

Hyperparameter	Value
Image size	224×224
Batch size	32
Max epochs	100
Learning rate	10^{-4} (Adam)
Latent channels C_z	64
$\beta_{\max}$	0.05
KL warm-up epochs	20
Initialisation	Random (no pre-training)
Anomaly threshold quantile	97.5%
Reconstruction weight	0.6
Random seed	42

3.5. Method Comparison

Table 4 places the two diagnostic approaches side by side on the four axes that matter in practice: input representation, training label requirement, model type, and interpretability output.

Table 4: Side-by-side comparison of diagnostic approaches. ✓: required; –: not required.

Approach	Paradigm	Input representation	Model	Labels?	Interpretability output
SpecRas + ResNet18 (SWA)	Supervised	1 200-dim PSD raster image	ResNet18 + SWA	✓	Grad-CAM per-Hz heat map
HetSpatVAE (SpatialResNet-VAE)	One-class detection	3-ch CWT/ STFT/context image	β-VAE, spatial latent	–	Reconstruction residual heat map

SpecRas preserves the full spectral shape in a 1 200-bin PSD-dominant image at the cost of a labelled training set and sensitivity to cross-unit covariate shift. HetSpatVAE replaces the discriminative objective with a generative one: learn to reconstruct normal time-frequency images, then flag anything that reconstructs poorly. No fault labels are required, and the spatial latent map localises the anomaly to specific CWT/STFT cells.

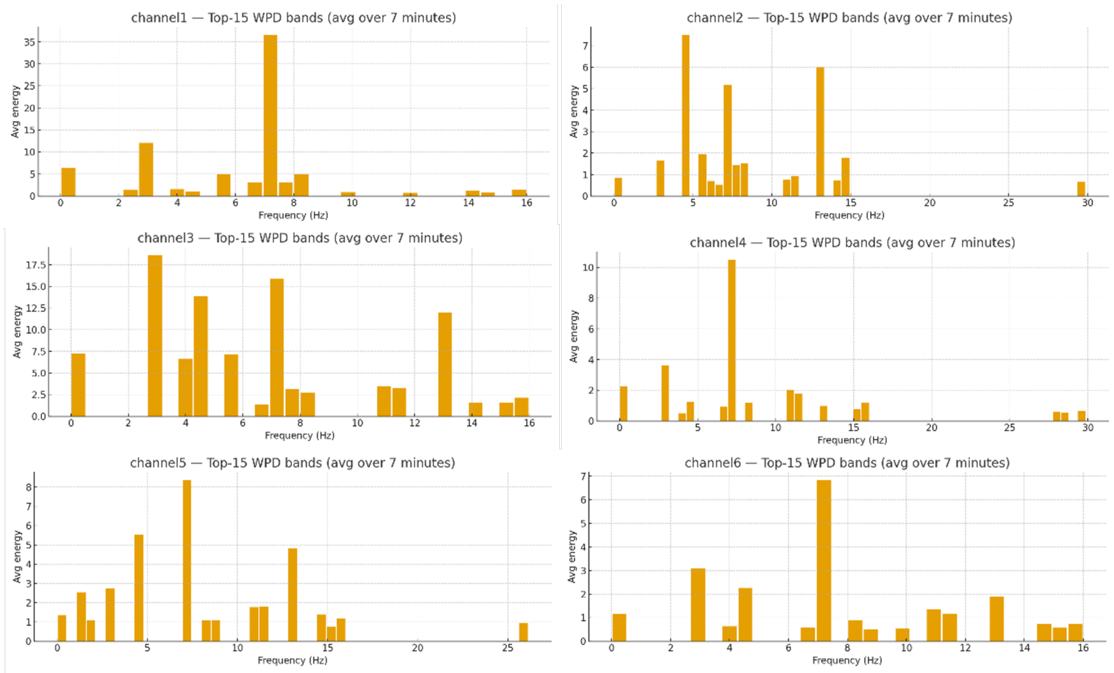
4. Experiments and Discussion

4.1. Dataset

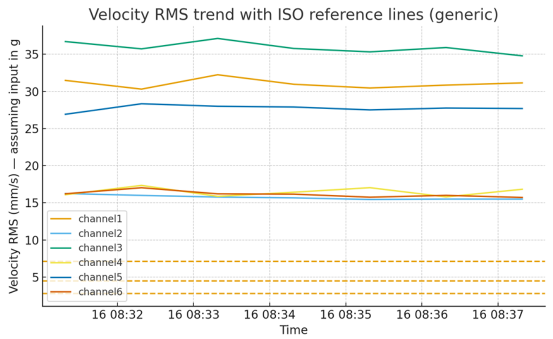
Vibration measurements were collected from real in-service AC power transformers at a regional substation. Each monitoring session is stored as a JSON file in which all records sharing a common timestamp t_k constitute one Segment. Each record carries a `sensor_id` string and a fixed-length `signal_value` array of $T = 8{,}192$ samples at $f_s = 8{,}192$ Hz (one second per Segment). No sensor position, orientation, or type metadata is stored in the collected data; the number of active channels U_k per Segment varies across sessions and transformer units. Energy-weighted channel fusion (Section 3.2) therefore collapses whatever channels are present into a single representative signal, requiring no knowledge of sensor placement or unit type. Table 5 summarises the train/validation/test split. Fig. 3 illustrates representative multi-channel signal characteristics at an in-service substation, motivating the energy-weighted fusion and the high-dimensional spectral feature schema.

Table 5: Dataset composition across 19 transformer units (4,499 Segments total). Each unit is assigned to exactly one split; no unit crosses split boundaries. Test units 134 and 135 were fully withheld from all development phases.

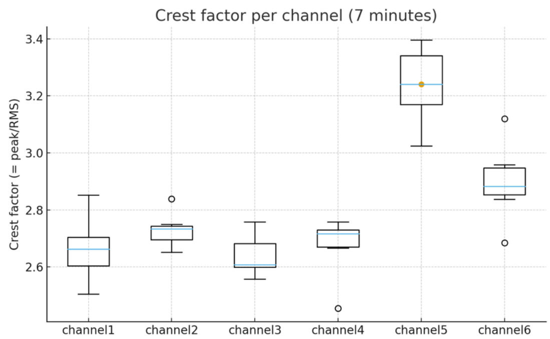
Split	Normal	Fault	Total	Transformer IDs
Train	1,595	2,249	3,844	120, 127–129, 143, 152 (normal); 114, 132, 136, 153–155, 157–159 (fault)
Validation	91	164	255	146 (normal); 145 (fault)
Test	246	154	400	**134 (normal); 135 (fault)**



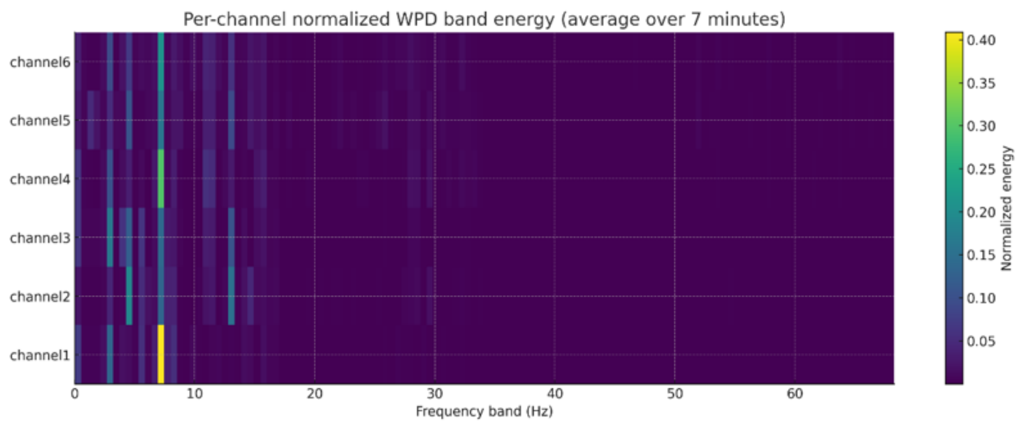
(a) Top-15 WPD frequency-band energy per sensor (7-min avg). Energy concentrates below 10 Hz; inter-channel levels vary substantially across all six sensors.



(b) Velocity RMS vs. time with ISO 10816 reference lines. Channels 1 and 3 exceed the upper caution threshold; channels 4–6 lie in the normal band.



(c) Crest factor (peak/RMS) per channel. Channels 5–6 show markedly elevated impulsive content (>3.2) versus 2.65–2.75 for channels 1–4.



(d) Per-channel normalised WPD heatmap (7-min avg). All channels concentrate energy in the 0–10 Hz sub-band; channels 5–6 show secondary excitation at 10–20 Hz.

Figure 3: Multi-channel signal characteristics (representative in-service substation). **(a)** WPD band energy: energy concentrated below 10 Hz, heterogeneous across 6 channels. **(b,c)** Velocity RMS trend and crest-factor distribution; channels 5–6 breach caution thresholds. **(d)** Normalised WPD heatmap confirming low-frequency dominance. Inter-channel heterogeneity motivates energy-weighted fusion (Eq. 2).

All 19 transformer units are assigned to exactly one split: 15 to training (6 normal-condition units and 9 fault-condition units), 2 to validation (units 146 and 145), and 2 to the held-out test set (units 134 and 135). This hard unit-level partition means that test accuracy measures whether the learned spectral representation transfers to transformers the model has never seen, not whether it interpolates within units already in the training pool. Test units 134 and 135 were not touched during feature design, hyperparameter search, or threshold calibration, making them a genuine out-of-sample evaluation.

4.2. Implementation Details

All experiments ran on a single workstation with an NVIDIA GPU (CUDA 12.8), Python 3.11, PyTorch 2.7+cu128, and torchvision 0.22. Scientific computing used NumPy 2.3, SciPy 1.16, and scikit-learn 1.5. The random seed was fixed at 42.

SpecRas training uses the hyperparameters in Table 2; training-set normalisation statistics for image encoding are computed once from the training split and frozen. HetSpatVAE training uses only normal-labelled training samples; the Mahalanobis statistics and anomaly threshold are computed from training-normal embeddings after the VAE converges.

4.3. Comparison Methods

Three diagnostic configurations are evaluated (two pathways, with two checkpoints for SpecRas):

1. **SpecRas + ResNet18 (best-val checkpoint):** the supervised pathway using the checkpoint with highest validation F1.
2. **SpecRas + ResNet18 (SWA checkpoint):** the same pipeline with the stochastic weight-averaged model.
3. **HetSpatVAE (SpatialResNetVAE):** the one-class anomaly detection pathway trained without fault labels, evaluated using the 97.5th-percentile anomaly threshold.

4.4. Performance Comparison

Table 6 reports five diagnostic metrics on the 400-sample held-out test set. Fig. 4 shows the HetSpatVAE visual diagnostics.

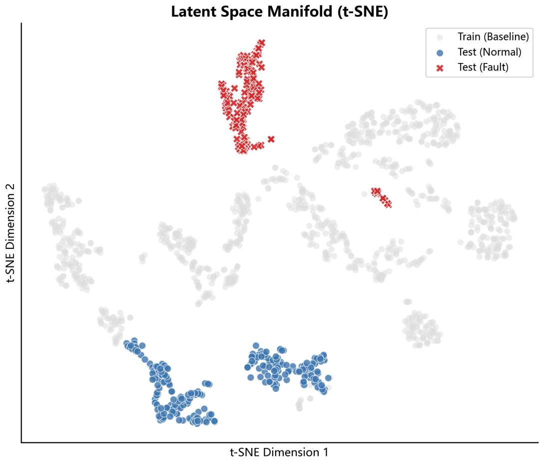
Table 6: Diagnostic performance on the held-out test set (246 normal / 154 fault, $N = 400$; transformers 134 and 135). Pre.(N): normal-class precision. Rec.(N/F): normal/fault recall. Best result per metric shown in **bold**. †: trained without fault labels.

Method	Paradigm	Acc.↑	Pre.(N)↑	Rec.(N)↑	Rec.(F)↑	Macro-F1↑
SpecRas + ResNet18 (best-val ckpt)	Supervised	94.25%	1.000	0.907	**1.000**	0.941
SpecRas + ResNet18 (SWA)	Supervised	97.00%	**1.000**	0.951	**1.000**	0.969
HetSpatVAE (SpatialResNetVAE)†	One-class	**98.25%**	**1.000**	**0.972**	**1.000**	**0.982**

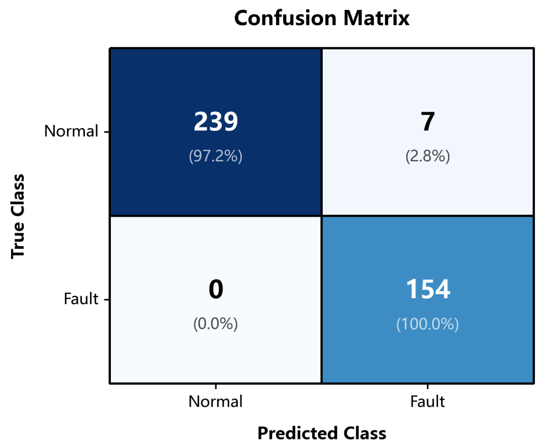
Fault recall. Both SpecRas checkpoints and HetSpatVAE achieve fault recall of 1.000: all 154 fault samples are correctly flagged. Zero false negatives across both deep-learning pathways is the operationally important result: a missed fault can lead to thermal runaway and equipment loss.

SWA narrows the false-positive gap. The SWA checkpoint reduces the false-positive count from $23/246 = 9.3\%$ to $12/246 = 4.9\%$ (normal recall $0.907 \to 0.951$). Each false positive triggers a field inspection that costs operational time and disrupts scheduled maintenance. The result is consistent with the flat-minima hypothesis (Izmailov et al., 2018): a wider loss basin generalises better across the covariate shift between training and test transformer units.

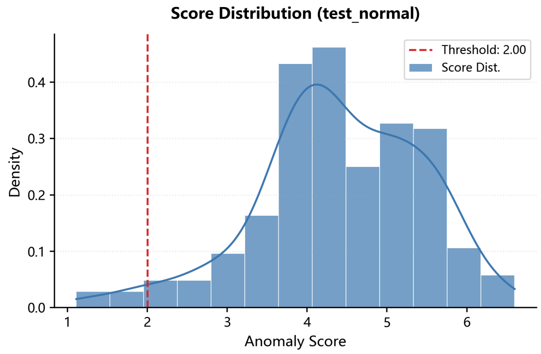
HetSpatVAE performance. Trained exclusively on normal samples, HetSpatVAE reaches 98.25% accuracy and macro-F1 of 0.982, surpassing the SWA supervised result by 1.25 pp, with 7 false positives (2.8% of normal). The t-SNE projection in Fig. 4(a) shows that test fault samples are systematically displaced from the training-normal cluster, confirming that the spatial VAE has learned a coherent normality manifold. Score histograms in Fig. 4(c,d) reveal the two well-separated score distributions: fault scores peak near 0.7 and normal scores above 3.5, with the threshold at $\tau = 2.00$ cleanly between the two modes. The confusion matrix (Fig. 4(b)) confirms 239 TN, 7 FP, 0 FN, 154 TP.



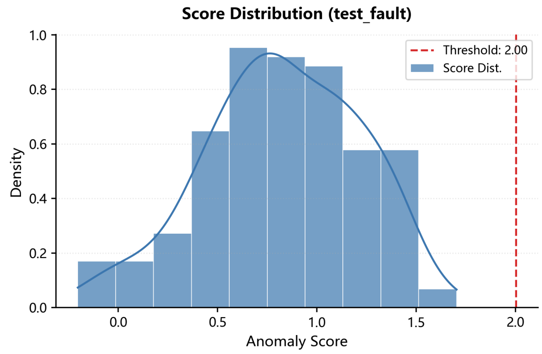
(a) Latent space (t-SNE): training normal (grey), test normal (blue, ∘), test fault (red, ×). Fault samples displaced from the normal cluster confirm a coherent normality manifold.



(b) Confusion matrix: 239 TN, 7 FP, 0 FN, 154 TP. Accuracy = 98.25%; fault recall = 1.000 (zero missed faults).



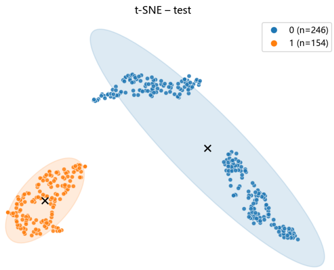
(c) Score distribution, test-normal ($n = 246$, $\tau = 2.00$). Mode near 4.0; 7 samples fall below τ (false-positive rate 2.8%).



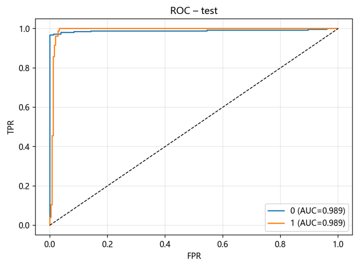
(d) Score distribution, test-fault ($n = 154$, $\tau = 2.00$). All fault scores lie below τ (mode $\approx$0.7): 100% fault recall.

Figure 4: HetSpatVAE diagnostic outputs on the held-out test set (τ at 97.5th training-normal percentile). **(a)** t-SNE: fault latent codes occupy a distinct region. **(b)** Confusion matrix: 98.25% accuracy, zero missed faults. **(c,d)** Score histograms: fault and normal distributions are well separated at $\tau = 2.00$.

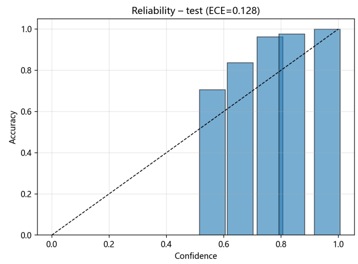
ROC, precision–recall, and calibration (SpecRas SWA). Fig. 5 shows the full diagnostic outputs for the SpecRas SWA checkpoint. The ROC AUC reaches 0.989 for both classes; the fault-class average precision (AP) is 0.960, so the ranked probability outputs are reliable regardless of which threshold is chosen. The reliability (calibration) curve gives an expected calibration error (ECE) of 0.128: a model output of 90% fault probability corresponds to an observed fault rate close to 90%. The t-SNE projection of penultimate-layer features shows two well-separated clusters, and the per-band PSD mean$\pm\sigma$ plots show fault samples carrying elevated spectral power at specific mid-to-high frequency bins.



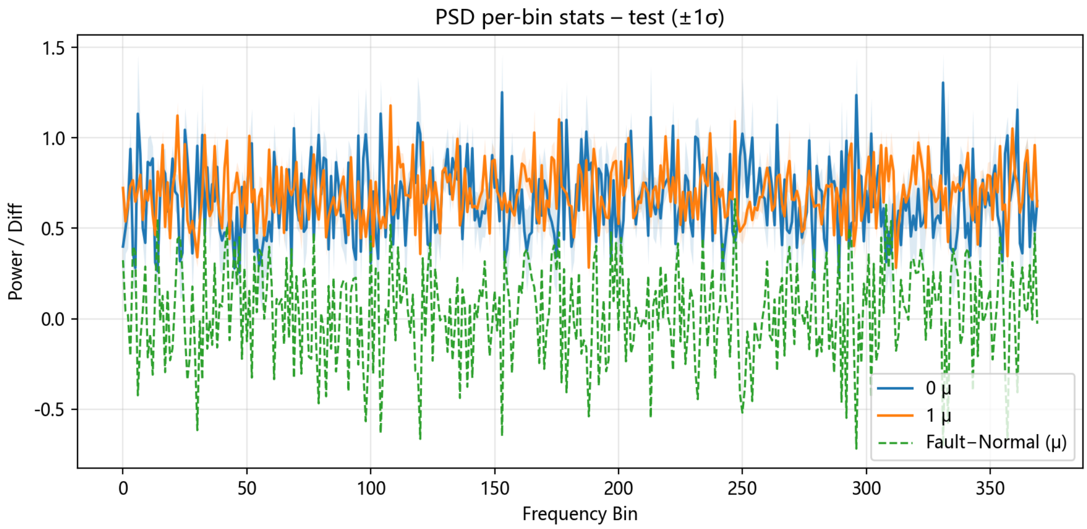
(a) t-SNE: normal (blue, $n = 246$) and fault (orange, $n = 154$) occupy disjoint clusters with non-overlapping confidence ellipses.



(b) ROC curves: AUC = 0.989 for both classes. TPR ≈ 0.97 at FPR = 0.02.



(c) Reliability diagram: ECE = 0.128. Bars track the diagonal; model outputs are conservative (slight under-confidence).



(d) PSD mean $\pm 1\sigma$ per frequency bin. Fault (orange) carries elevated power at mid-to-high frequency bands; Fault−Normal difference (green dashed) marks the discriminative Hz regions.

Figure 5: SpecRas SWA checkpoint (97.00% accuracy, macro-F1 = 0.969; 234 TN, 12 FP, 0 FN, 154 TP). **(a)** t-SNE feature clusters. **(b)** ROC (AUC = 0.989). **(c)** Calibration (ECE = 0.128). **(d)** PSD statistics: fault samples show elevated spectral power in specific bands identified by the difference curve.

Validation performance. Both SpecRas checkpoints achieve 100% accuracy and F1 on the validation set ($N = 255$). The gap to test accuracy (97%) is attributed entirely to cross-unit covariate shift, not overfitting.

4.5. Ablation: SWA Contribution

Table 7 isolates SWA by comparing best-val and SWA checkpoints, which differ only in the weight-averaging step.

Table 7: Ablation of stochastic weight averaging (SWA).

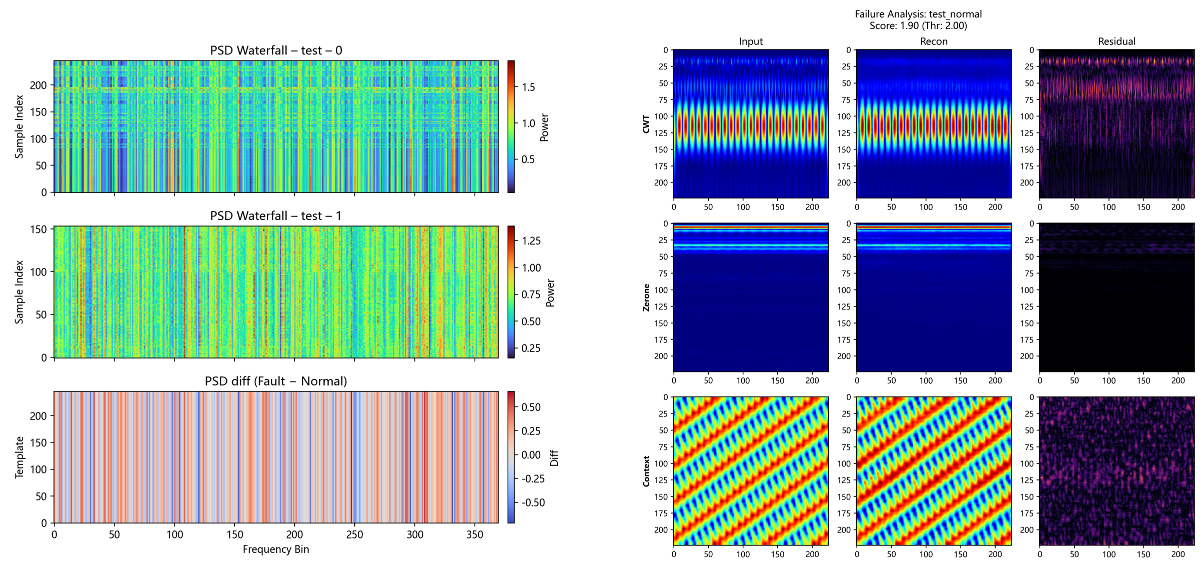
Configuration	Acc.↑	Macro-F1↑
SpecRas + ResNet18 (best-val ckpt)	94.25%	0.941
SpecRas + ResNet18 (SWA ckpt)	**97.00%**	**0.969**

SWA contributes +2.75 pp accuracy and +0.028 macro-F1 at zero additional inference cost.

4.6. Misclassification and Visualisation Analysis

All 12 false positives from the SpecRas SWA checkpoint originate from transformer 134 and cluster in the 16:00–20:30 local-time window (peak evening load). During this period, magnetostriction-induced broadband excitation is elevated, pushing high-frequency spectral content into the range that overlaps genuine fault signatures. The HetSpatVAE false positives (7 samples) overlap the same time window and show elevated reconstruction error concentrated in the 1 000–2 000 Hz STFT channel—channel-level root-cause evidence consistent with the SpecRas Grad-CAM attribution maps.

Spectral fingerprint. Fault samples carry a distinctive PSD pattern: a dominant peak near $2f_0 = 100\,\mathrm{Hz}$ and secondary clusters at 300 Hz and 500 Hz (odd harmonics of winding resonance under magnetostriction). Grad-CAM activations on `layer4` consistently highlight the corresponding row bands in the PSD panel (Fig. 2). Fig. 6(a) confirms this pattern in the PSD waterfall: the per-sample Fault−Normal difference is broadly positive across 50–300 Hz, with localised peaks at the harmonic frequencies. Fig. 6(b) dissects a representative HetSpatVAE false positive whose score (1.90) falls just below the threshold (2.00): reconstruction residuals are largest in the CWT channel at the 100 Hz row, consistent with a peak-load magnetostriction harmonic rather than a genuine fault signature.

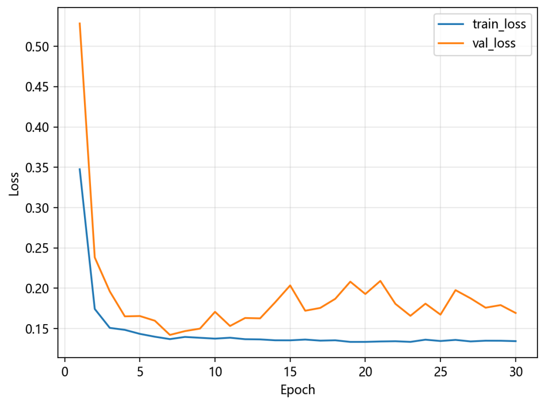


(a) PSD waterfall (test set). Top/middle: per-sample normalised PSD for normal/fault. Bottom: Fault−Normal difference; red bands mark elevated power in the 50–300 Hz harmonic region.

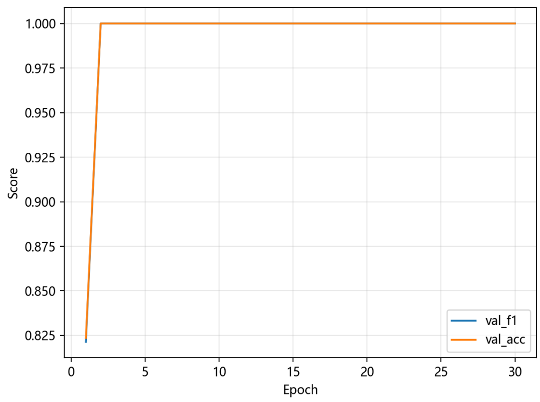
(b) HetSpatVAE false positive (score = 1.90, $\tau = 2.00$). Columns: input / reconstruction / residual. Rows: CWT / SpecRas raster / waveform context. Residuals concentrate at the 100 Hz CWT row (peak-load magnetostriction, not a structural fault).

Figure 6: Spectral fingerprint and false-positive analysis. **(a)** PSD waterfall: Fault−Normal difference highlights 50–300 Hz harmonic bands; Grad-CAM hot rows in the PSD panel correspond to the same frequencies. **(b)** VAE false-positive residuals localise to the 100 Hz CWT row—peak-load magnetostriction excitation, not a fault.

Training dynamics. The ReduceLROnPlateau schedule brings the learning rate from 8×10^{-6} to 8×10^{-8} over 30 epochs. Validation accuracy and macro-F1 both reach 1.000 by epoch 2 and remain there, reflecting clean spectral separation between training normal and fault units. The 3 pp drop from validation (100%) to test (97%) comes from cross-unit covariate shift—transformer 134 was recorded under a different time-of-day distribution, not a harder fault signature. Fig. 7 shows the training and validation loss curves together with the validation accuracy and macro-F1 tracks.



(a) Loss vs. epoch. Train converges to ≈0.14 by epoch 5; val stabilises at ≈0.17 (cross-unit shift, not overfitting). SWA accumulates from epoch 15 over the flat region.



(b) Validation accuracy and macro-F1 vs. epoch. Both metrics saturate at 1.000 from epoch 2; the flat plateau enables stable SWA accumulation.

Figure 7: SpecRas ResNet18 training dynamics (30 epochs). **(a)** Loss: rapid convergence; val plateau enables SWA. **(b)** Val accuracy and F1 saturate at 1.000 by epoch 2.

Fig. 1 shows the dual-pathway architecture and how energy-weighted channel fusion feeds each diagnostic branch. Fig. 2 diagrams the raster grid: the PSD panel (≈77% of image height) dominates the layout, and a Grad-CAM hot region at panel row r, column c maps to $1 + 32r + c$ Hz. Fig. 3 illustrates representative multi-channel vibration characteristics motivating the energy-weighted fusion. Fig. 7 shows the SpecRas training dynamics: loss convergence by epoch 5 and validation-metric saturation by epoch 2, enabling stable SWA accumulation from epoch 15. Fig. 5 collects the SpecRas diagnostic outputs: t-SNE feature clusters, ROC curves (AUC = 0.989), calibration curve (ECE = 0.128), and per-band PSD statistics. Fig. 6 presents the PSD waterfall and a representative HetSpatVAE false-positive reconstruction breakdown, localising the residual to the peak-load 100 Hz CWT row.

4.7. Discussion

Three findings stand out.

Why spectral representation matters. A model that sees the full shape of the PSD—1 000 bins rather than aggregate scalars—can distinguish a peak-load harmonic shift from a sustained fault signature, whereas coarser features cannot. The raster encoding makes this tractable for a CNN: with 77% of the image area devoted to PSD content, most receptive fields land on spectral data, and the row-major wrap preserves frequency ordering so Grad-CAM attribution reads directly as Hz-resolution spectral bands.

One-class detection versus supervised classification. HetSpatVAE outperforms the supervised SWA checkpoint by 1.25 pp (98.25% vs. 97.00%). The spatial VAE optimises a generative target: reconstruct normal time-frequency images well enough that anything anomalous stands out by its reconstruction error. A generative model trained on normal data alone generalises more reliably to unseen transformer units than a discriminative classifier fitted on both classes. The score histogram in Fig. 4(c,d) also shows why the composite score matters: fault and normal distributions are clearly separated at $\tau = 2.00$, with the normal-class peak at $\approx$4.0 well above the fault-class peak at $\approx$0.7. The false-positive reconstruction analysis (Fig. 6(b)) shows that the 7 misclassified normal samples have residuals concentrated at the 100 Hz CWT row—a peak-load harmonic, not a fault signature, and one that the composite reconstruction-plus-Mahalanobis score keeps just below threshold.

Load conditions as a confounding variable. All false positives from both pathways fall in the 16:00–20:30 peak-load window. One direct remedy is to condition predictions on ambient load level by including the transformer tap-changer position or bus voltage as a covariate. Alternatively, a time-of-day calibration that widens the normal-class boundary during known peak-load periods would address it without requiring additional sensor data.

Computational overhead. Feature extraction (SpecRas) processes approximately 4 000 samples per CPU-hour. ResNet18 training completes in under 30 minutes (30 epochs, batch size 16, GPU). HetSpatVAE training takes approximately 2 hours (50 epochs, batch size 32, GPU). Online inference runs in under 100 ms per Segment, fast enough for near-real-time monitoring.

5. Conclusion

Three things make vibration-based transformer fault diagnosis genuinely difficult: the signals change with load, confirmed fault examples are rare, and engineers will not act on a black-box score alone. This paper addressed all three simultaneously.

The SpecRas supervised pathway encodes a 1 200-dimensional physics-motivated spectral feature vector as a raster-stripe image and trains a ResNet18 classifier. The SWA checkpoint reaches 97.00% accuracy and macro-F1 of 0.969 on a cross-unit held-out test set, with zero missed faults. The +2.75 pp gain from SWA over the best-validation checkpoint confirms that flat-loss-basin regularisation is worth applying whenever training data are limited and cross-unit covariate shift is expected.

The HetSpatVAE pathway trains a SpatialResNetVAE on normal-only data, retaining a 64×7×7 spatial latent map that produces reconstruction-residual heat maps over the time-frequency plane. On the same test set it reaches 98.25% accuracy and macro-F1 of 0.982 without using a single fault label. In practice, substations can go years without a confirmed fault, so a label-free detector can be deployed immediately rather than waiting for labelled failures to accumulate.

All false positives from both pathways cluster in the evening peak-load window and share elevated 1 000–2 000 Hz STFT content, pointing to a physical rather than a modelling cause. Incorporating load metadata—tap-changer position or bus voltage—is the most direct corrective step.

Limitations and future work. The current evaluation covers two transformer units at one substation sharing a common operating schedule. Transfer to units with different core designs (three-phase shell-type vs. core-type), voltage ratings, or 60 Hz grids remains an open question; federated or domain-adaptive transfer learning is the natural approach. All fault conditions are grouped into a single class—fine-grained classification of winding deformation, core looseness, and cooling faults requires labelled examples per fault type, which HetSpatVAE residual heat maps can help accumulate by pointing engineers to candidate anomaly events. Finally, Morlet CWT at 128 scales takes roughly 30 ms per sample on a modern CPU, too slow for microcontroller-based sensor nodes; a learnable filter-bank approximation would address this.

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Data availability

The vibration dataset was collected under a confidentiality agreement with the grid operator and cannot be made publicly available.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used ChatGPT (OpenAI) in order to polish the writing and to check grammar and spelling. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Algorithm 1 SpecRas Feature Extraction

Require: Multi-channel signals $\mathcal{X}_k$, sampling rate f_s, feature schema $\mathcal{S}$
Ensure: Feature vector $\mathbf{S}_k \in \mathbb{R}^{1200}$

Compute channel weights w_u via Eq. (2)
for each channel $u = 1, \dots, U_k$ **do**
  Remove DC: $\bar{x}^{(u)} \leftarrow x_k^{(u)} - \mu_{x_k^{(u)}}$
  Extract time features $\mathbf{s}_{\text{time}}^{(u)}$ (15 dims)
  Compute Welch PSD via Eq. (8)
  Extract STFT means $\mathbf{s}_{\text{stft}}^{(u)}$ (127 dims)
  Extract PSD features $\mathbf{s}_{\text{psd}}^{(u)}$ (1 050 dims)
  Extract HF ratios $\mathbf{s}_{\text{hf}}^{(u)}$ (8 dims)
  Concatenate: $\mathbf{s}^{(u)} \leftarrow [\mathbf{s}_{\text{time}}^{(u)}; \mathbf{s}_{\text{stft}}^{(u)}; \mathbf{s}_{\text{psd}}^{(u)}; \mathbf{s}_{\text{hf}}^{(u)}]$
end for
Aggregate: $\mathbf{S}_k \leftarrow \sum_u w_u \mathbf{s}^{(u)}$ via Eq. (12)
Normalise via Eq. (14)
return $\mathbf{S}_k$

Appendix A. Feature Extraction Algorithm

Algorithm 1 summarises the SpecRas feature extraction pipeline for a single multi-channel sample.